If the given lines intersect, then there must be a common point. Therefore, for some $s, t \in \mathbb{R}$, we have $(2s+1,\ 3s+3,\ 2s-1) = (t+2,\ 2t+4, 4t-3)$.

Equating the coordinates of x, y and z we get

$$2s - t = 1,\ 3s - 2t = 1 \text{ and } s - 2t = -1.$$

Solving the first two of the above three equations, we get $s = 1$ and $t = 1$. These values of s and t satisfy the third equation. So, the lines are intersecting.

Now, using the value of s in (1) or the value of t in (2), the point of intersection $(3,6,1)$ of these two straight lines is obtained.

If we take $\vec{b} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{d} = \hat{i} + 2\hat{j} + 4\hat{k}$, then $\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{vmatrix} = 8\hat{i} - 6\hat{j} + \hat{k}$ is a vector perpendicular to both the given straight lines. Therefore, the required straight line passing through $(3,6,1)$ and perpendicular to both the given straight lines is the same as the straight line passing through $(3,6,1)$ and parallel to $8\hat{i} - 6\hat{j} + \hat{k}$. Thus, the equation of the required straight line is

$$\vec{r} = (3\hat{i} + 6\hat{j} + \hat{k}) + m(8\hat{i} - 6\hat{j} + \hat{k}),\ m \in \mathbb{R}.$$

Example 6.35

Determine whether the pair of straight lines $\vec{r} = (2\hat{i} + 6\hat{j} + 3\hat{k}) + t(2\hat{i} + 3\hat{j} + 4\hat{k})$, $\vec{r} = (2\hat{j} - 3\hat{k}) + s(\hat{i} + 2\hat{j} + 3\hat{k})$ are parallel. Find the shortest distance between them.

Solution

Comparing the given two equations with

$$\vec{r} = \vec{a} + s\vec{b} \text{ and } \vec{r} = \vec{c} + s\vec{d},$$

we have $\vec{a} = 2\hat{i} + 6\hat{j} + 3\hat{k},\ \vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k},\ \vec{c} = 2\hat{j} - 3\hat{k},\ \vec{d} = \hat{i} + 2\hat{j} + 3\hat{k}$

Clearly, $\vec{b}$ is not a scalar multiple of $\vec{d}$. So, the two vectors are not parallel and hence the two lines are not parallel.

The shortest distance between the two straight lines is given by

$$\delta = \frac{|(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}$$

$$\text{Now, } \vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 2 & 3 \end{vmatrix} = \hat{i} - 2\hat{j} + \hat{k}$$

$$\text{So, } (\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = (-2\hat{i} - 4\hat{j} - 6\hat{k}) \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 0.$$

Therefore, the distance between the two given straight lines is zero.Thus, the given lines intersect each other.

Chapter 6 Vector Algebra.indd 253 31-01-2020 17:52:01